



## **ME329 Group 1-5 Gearbox Mechanical Design Report**



Figure 1. Gearbox Assembly.

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## **Abstract/Intro:**

This project involved the design, modification, analysis, and testing of a compact gearbox capable of supporting suspended loads while meeting specified cost, size, and performance requirements. The final design utilized a modified Harbor Freight hand winch with a 40:1 worm gear reduction system, a custom-machined output shaft, and a simplified fastening arrangement. Engineering analyses were performed to evaluate shaft stresses, worm gear performance, deflection, and the strength of critical fastening components. Results showed that the worm gear provided effective torque multiplication, requiring only 2.67 lbf of user input force while maintaining self-locking capability to prevent back-driving. Testing demonstrated that the gearbox successfully supported loads up to 170 lb, with a maximum limit test of 300 lb before failure due to set screw slip. The analysis identified the shaft keyway as a potential critical location; however, testing revealed the set screw connection to be the governing failure mode. With a total cost of only \$56, the design achieved the highest performance-per-dollar ratio among all groups. Overall, the project successfully met its objectives and demonstrated the application of mechanical design, manufacturing, and testing principles in the development of a functional gearbox system.

## **Design Overview and Relevant Background:**

### **Problem Statement and Design Goal:**

For this project, the objective was to design, manufacture, and test a compact gearbox capable of supporting a suspended load while satisfying specified performance, size, and cost constraints. The gearbox was required to provide sufficient torque multiplication so that a user could operate the system with minimal input force while safely resisting back-driving under load. Additional design requirements included maintaining a factor of safety greater than one, minimizing deflection, fitting within dimensional constraints, and keeping overall manufacturing costs low. The final design goal was to develop a reliable, low-cost gearbox that could efficiently transmit power, safely support loads, and demonstrate sound mechanical engineering design principles through analysis, fabrication, and experimental testing.

### **How it Works:**

The gearbox operates using a worm-and-gear reduction system adapted from a commercial hand winch. When the user rotates the input handle, the worm gear drives a 40-tooth spur gear attached to the output shaft. This large gear reduction increases the available output torque while significantly reducing the force required from the user. The output shaft transfers torque to the weight attachment fixture, allowing external loads to be lifted or supported. Because the worm gear has a small lead angle, the mechanism is self-locking, preventing the load from back-driving the gearbox when the handle is released. This feature improves safety and allows the system to maintain a load without continuous user input.

## Design Constraints and Criteria:

To ensure the gearbox is functional, the following specifications were established:

The primary challenges involved adapting commercial components to work with custom-machined parts while maintaining proper fit and function. Modifications to the winch spool, custom shaft, and handle required careful machining and repeated test fitting to achieve the desired tolerances. During operation, the design needed to balance torque capacity, structural strength, and ease of use while remaining within the project's size and cost limits. Testing ultimately revealed that the set screw connection between the shaft and spool was the weakest component, highlighting the importance of properly designing power-transmitting joints in addition to the primary structural members.

Table 1. Design Criteria and Target Specifications

| Design Criteria  | Specification |
|------------------|---------------|
| Weight Capacity  | > 50 lbs      |
| Target RPM       | ~ 3-4 RPM     |
| Size             | < 18x18x12 in |
| Factor of Safety | >1            |

## Concepts and Ideation:

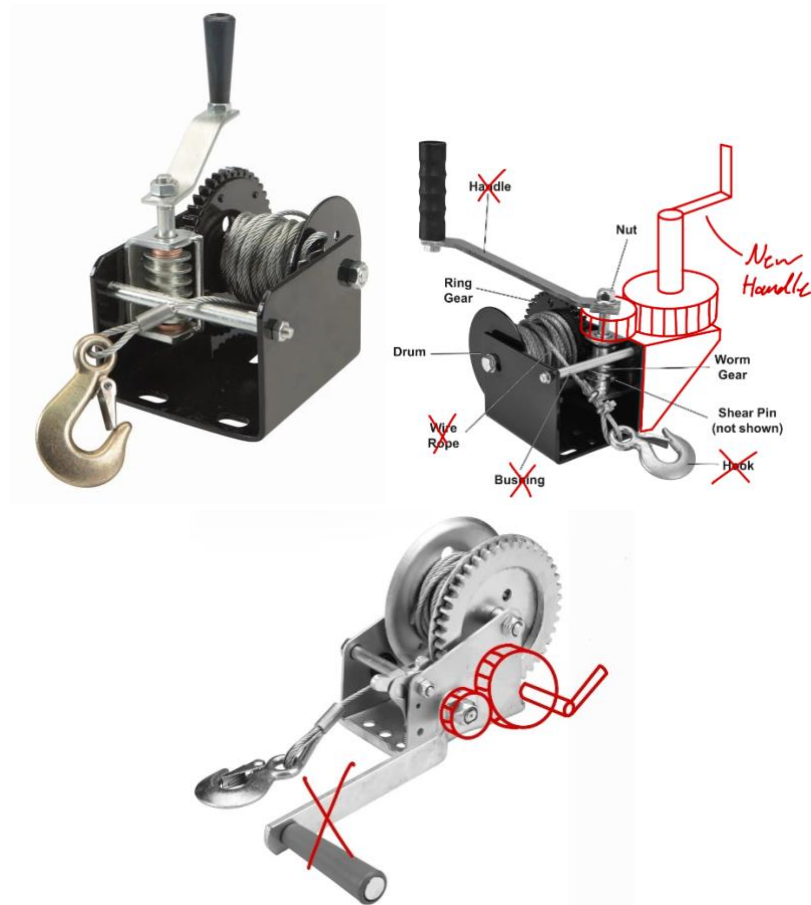


Figure 2. Gearbox Concepts.

As shown in Figure 2, we brainstormed three initial design concepts. The top left concept was selected because it provided the best combination of mechanical advantage, compact size, low cost, and ease of manufacturing. The worm gear configuration also offered self-locking capability, improving safety and preventing back-driving under load.

## **Construction, Operation, and Safety Challenges and Concerns:**

### **Construction:**

We began with a 2000 lb Harbor Freight Hand Winch. We modified the product extensively, beginning with a complete disassembly, leaving only the worm gear and its housing attached to the steel frame.

We enlarged the hole in the spur gear spool to fit our larger designed shaft of 5/8". Additionally, we drilled and tapped a 1/4 - 20 threaded hole into the hub of the spool. This would fit a set screw to fix the spool to the shaft.

The shaft started as a 5/8" steel rod. It required a few skim passes to fit properly with the gear spool. Additionally, we turned the last inch and a half down to .495" to fit the weight attachment fixture. This took a few attempts, going back and forth between machining and testing with the fixture to achieve the right fit. We didn't want to accidentally undersize the end of the shaft. We also machined a matching 1/8" keyway to meet with the weight fixture. Again, this took a few attempts, needing adjustments to compensate for our decrease in diameter.

Lastly, we modified the existing handle to shorten its overall length and fit within the desired space requirements. The handle was simply cut short on a bandsaw, with a hole drilled for an additional 1/2 inch bolt. This allowed for a much easier rotation and eliminated possible conflict with the weight arm.

### **Operation and Ergonomics:**

The gearbox was designed to be simple and intuitive to operate while minimizing the physical effort required from the user. The 40:1 worm gear reduction provided a substantial mechanical advantage, allowing relatively large loads to be supported with a small input force. Based on the gear analysis, only approximately 2.67 lbf was required at the handle under maximum design loading, making operation comfortable and well below the project guideline of 10 lbf.

To improve usability, the original winch handle was shortened to fit within the dimensional constraints of the project and to prevent interference with the weight arm during testing. Despite the reduced length, the handle still provided sufficient leverage for smooth operation. The self-locking nature of the worm gear also improved ergonomics by eliminating the need for the operator to continuously hold the handle to maintain a load. This allowed us to safely pause operation without concern for the load back-driving the mechanism.

### Safety Concerns:

Since the gearbox was designed to support suspended loads, ensuring the load remained securely supported was a primary safety consideration. The self-locking worm gear prevents the system from rotating backward under load, allowing it to hold weight safely when the handle is released. Potential failure modes included shaft yielding, gear failure, and set screw slip. Analysis identified the keyway as a potential weak point due to stress concentrations, while the showcase loading revealed that the set screw connection between the shaft and spool was the actual limiting component. To improve safety and reliability in future designs, a second set screw or a stronger method of securing the spool to the shaft should be implemented. This would reduce the likelihood of slipping and allow the assembly to safely support higher loads.

### Cost Breakdown:

Table 2. Itemized Cost of the Gearbox

| Store          | Items Purchased  | Price (\$) |
|----------------|------------------|------------|
| Home Depot     | Set Screws       | 1          |
|                | Shaft            | 6          |
|                | ½-13 5 Inch Bolt | 1.30       |
|                | ½-13 Nut x2      | .60        |
| Amazon         | Shaft Collars    | 7          |
| Harbor Freight | Hand Winch       | 40         |
| Total          |                  | 56         |

## Calculations:

Table 3: Material Properties of various assembly components

| Component                                    | Ultimate Strength<br>(kpsi) | Yield Strength<br>(kpsi) | Modulus of Elasticity<br>(psi) |
|----------------------------------------------|-----------------------------|--------------------------|--------------------------------|
| Shaft – 1045 HR                              | 82                          | 45                       | $30 \times 10^6$               |
| Spur Gear - Hardened<br>1008 Steel HR        | 45                          | 25                       | $30 \times 10^6$               |
| Worm Gear – Hardened<br>Carbon 1045 Steel HR | 82                          | 45                       | $30 \times 10^6$               |

## Shaft Calculations

1 Cantilever—end load

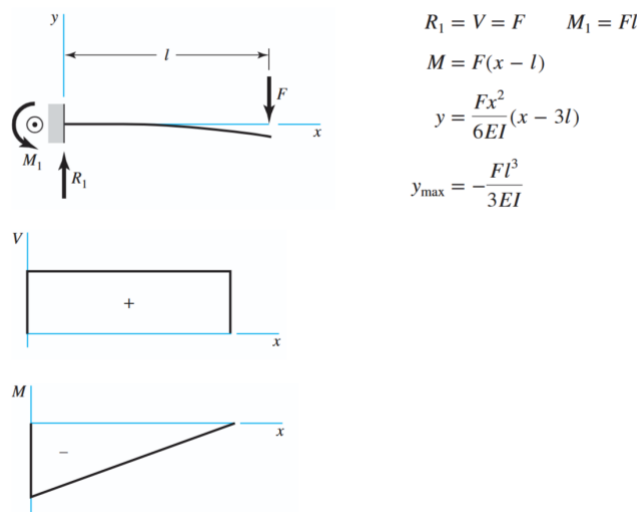


Figure 5. Schematic of Cantilever Load Case.

A fatigue and static yield analysis was performed on the critical section of the shaft ( $d=0.500$  in) to evaluate stress concentrations at a machined keyway and a shoulder step. The shaft is subjected to an alternating bending moment ( $M_a=62.5$  lb-in) and a mean torque ( $T_m=150$  lb-in), with a target design safety factor of  $n_{\text{design}} = 1.5$ . (Note: Full calculation parameters and MATLAB outputs are provided in the Addendum).

### **Stiffness and Deflection**

Based on a cantilever model with an effective length of 4.0 in:

- Maximum Deflection ( $y_{\max}$ ): 0.0036 in
- End Slope ( $\theta_{\max}$ ): 0.00136 rad

### **Fatigue Life (Modified Goodman)**

Using Marin modifying factors, the corrected endurance limit was determined to be  $S_e=15.34$  ksi. Applying notch sensitivities for bending and torsion:

- Keyway:  $n_{\text{key}}=0.88$  (*Governing*)
- Shoulder Step:  $n_{\text{step}}=0.92$
- Fatigue Result: Fail ( $0.88 < 1.5$ )

### **Static Yield (von Mises)**

Evaluating the maximum combined stresses against the material yield strength ( $S_y=25$  ksi):

- Governing Yield Factor:  $n_{\text{yield}}=1.585$  (at the key)

### **Conclusion and Recommendations**

The current shaft design meets the 1.5 safety factor requirement for static yielding, with fatigue failure predicted at the keyway. However, fatigue was determined to be a non-critical design factor. This could be fixed with higher quality steel such as 1045CD.

## **Worm Gear Calculations**

A kinematic and stress analysis was performed on the winch's worm gear reduction system to evaluate efficiency, self-locking capability, and strength under maximum testing loads. The system uses a worm ( $N_w=1$ ) and a 40-tooth gear ( $N_g=40$ ) for a 40:1 ratio, driving a maximum load of  $P_{\text{load}}=12.5$  lbf. (*Note: Full calculation parameters and MATLAB outputs are provided in the Addendum*).

### **Kinematics & Self-Locking**

- Lead Angle ( $\lambda$ ):  $6.09^\circ$
- System Efficiency: 39.37%
- Self-Locking Result: Yes (The friction coefficient of  $\mu = 0.15$  overcomes the static back-drive requirement). This ensures the winch is safe for static load holding.

### **Input Requirements**

- User Input Force: 2.67 lbf at the handle. This easily satisfies the  $<10$  lbf guideline.

### **Stress & Durability**

Evaluating hardened alloy steel against a required safety factor of  $n_{\text{safety}} = 2.0$ :

- Bending Stress: 3,375 psi (SF = 44.4)
- Contact Stress: 29,835 psi (SF = 5.0)

### **Conclusion**

The current worm gear design passes all requirements. The system efficiently steps down input force to a manageable level while maintaining a self-locking geometry to prevent back-driving under load.

## **Bolt Calculations**

A stress analysis was performed on the 1/2-13 steel bolt used in the handle assembly to verify that it could safely withstand the maximum expected operating loads. The bolt was evaluated under a conservative single-shear loading condition based on the maximum allowable user input force.

Bolt diameter:

$$d = 0.50 \text{ in}$$

Maximum user force:

$$F = 10 \text{ lbf}$$

### **Shear Stress**



Assuming the bolt is loaded in single shear:

$$A = \frac{\pi d^2}{4}$$
$$A = \frac{\pi(0.50)^2}{4} = 0.1963 \text{ in}^2$$
$$\tau = \frac{F}{A} = 50.9 \text{ psi}$$

### Factor of Safety

For low-carbon steel bolts:

$$S_y \approx 36,000 \text{ psi}$$

Using distortion energy theory:

$$S_{sy} = 0.577S_y$$
$$S_{sy} = 0.577(36,000) = 20,772 \text{ psi}$$
$$n = \frac{20,772}{50.9} = 408$$

### Conclusion:

The bolt experiences negligible stress compared to its allowable strength and greatly exceeds the required safety factor. Therefore, bolt failure is not considered a critical failure mode for the gearbox design.

## Set Screw Calculations

A torsional analysis was performed on the 1/4-20 set screw used to secure the spool to the output shaft. The set screw was evaluated under the maximum transmitted torque of 150 lb-in to determine its ability to prevent rotational slip.

Shaft diameter:

$$d = 0.625 \text{ in}$$

Applied torque:



$$T = 150 \text{ lb} \cdot \text{in}$$

The torque creates a tangential force:

$$F_t = \frac{T}{r}$$

where

$$r = \frac{d}{2} = 0.3125 \text{ in}$$
$$F_t = 480 \text{ lbf}$$

### Shear Stress in Set Screw

For a 1/4-20 screw:

Minor diameter:

$$d_r \approx 0.1887 \text{ in}$$

Shear area:

$$A = \frac{\pi d_r^2}{4}$$
$$A = 0.0280 \text{ in}^2$$

Shear stress:

$$\tau = \frac{480}{0.0280}$$
$$\tau = 17,143 \text{ psi}$$

### Factor of Safety

For a Grade 2 steel screw:

$$S_{sy} = 20,000 \text{ psi}$$

$$n = 1.17$$

**Conclusion:**

The set screw has the lowest factor of safety of any major component in the assembly and was identified as the governing failure point. This result agrees with showcase testing, where set screw slip occurred before any significant structural failure of the shaft or gearbox components.

**Shaft Collar Calculations**

An axial retention analysis was performed on the pair of 5/8-inch shaft collars used to position and retain the output shaft. The collars were evaluated using the maximum load achieved during testing to verify that they could safely resist shaft movement.

Maximum test load:

$$P = 300 \text{ lbf}$$

Number of collars:

$$N = 2$$

Load per collar:

$$F = \frac{300}{2} = 150 \text{ lbf}$$

**Commercial Rating Check**

Typical 5/8-inch steel shaft collars have a clamping force greater than:

$$F_{allow} = 1000 \text{ lbf}$$

Factor of safety:

$$n = \frac{F_{allow}}{F}$$
$$n = 6.67$$

### Conclusion:

The shaft collars provide more than adequate axial retention and exceed the required safety factor by a significant margin. No collar-related issues were observed during operation, indicating that the shaft collars are not a limiting component in the current design.

### Critical Points: Failure Modes

During the initial design phase, the shaft keyway was anticipated to be the primary point of failure due to the high stress concentrations at the machined keyway. However, tests of the system's load revealed that the set screw is the weak link. While the shaft does exhibit low safety factors under high loading, the set screw's limited force makes it vulnerable to the system's torque. Therefore, the governing failure mode shifts from shaft fatigue or yielding to set screw shear or torsional slip, meaning the mechanical joint is expected to fail long before the shaft experiences fracture.

### Testing Results:

Table 4: Testing Results Summary for All Groups

| Group #  | Test 1     | Test 2                                            | Test 3             | Cost (\$) | Limit Test                      |
|----------|------------|---------------------------------------------------|--------------------|-----------|---------------------------------|
| 1        | 53         | 225<br>(backswing)                                | 385 (1/3)          | 140       |                                 |
| 2        | 106        | 202                                               | 294 (pin<br>shear) | 120       |                                 |
| 3        | 54         | <del>290</del> -(housing)                         | N/A                | 180       |                                 |
| <b>5</b> | <b>106</b> | <b>214-Fail to<br/>incorrect key<br/>mounting</b> | <b>170</b>         | <b>56</b> | <b>300 (Set<br/>Screw Slip)</b> |
| 6        | 54.25      | 224.25-(Pin<br>Shear)                             | 99                 | 160       |                                 |

An analysis of the testing data demonstrates that our design (Group 5) achieved the strongest performance per dollar among all teams. With a total manufacturing cost of just \$56, less than half the cost of the next most economical design, we maintained the highest overall efficiency. During testing, our assembly successfully held a valid load of 170. Our subsequent limit test reached a load of 300; however, this run is invalid because the assembly failed due to a set screw slip. As a result, our true operational limit falls somewhere between 170 and 300. Because this failure was a fastening limitation rather than a catastrophic structural yield of the core components, it is highly

probable that our actual mechanical limit closely aligns with Group 2's data. Group 2 experienced a critical pin shear at a load of 294; had our set screw held to allow us to find the real limit, we anticipate our core design would have reached a similar threshold before shaft failure. From this testing, our next course of action would be adding a second set screw which is an iteration that requires little additional work and cost.

## **Recommendations for Final Full Scale Version:**

To optimize the final full-scale version, we recommend several general enhancements alongside two critical structural improvements. General upgrades, such as adding ball bearings and double supports to eliminate bending moments, utilizing a custom worm gear for the correct RPM, and incorporating a mounting base and an ergonomic handle, will greatly improve overall efficiency and usability. However, the design would become significantly more robust by directly addressing our primary testing bottleneck. Upgrading to a higher-grade alloy, like 1045 Cold Drawn (CD) steel, will drastically increase overall strength. Furthermore, because our limit test ultimately failed due to a fastening slip, adding a second set screw is needed to substantially increasing the system's torsional holding power. A hardened shaft with a dual-set-screw configuration would resolve the assembly's weakest link, prevent rotational slip and allow the design to have higher mechanical limits.

## **Conclusion and Reflection:**

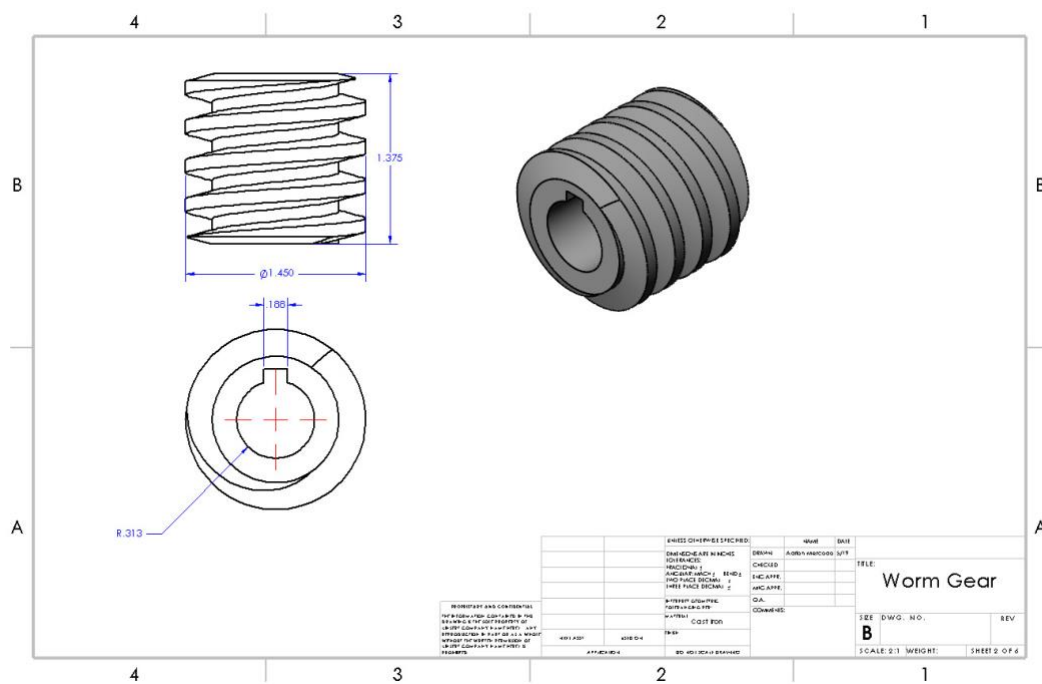
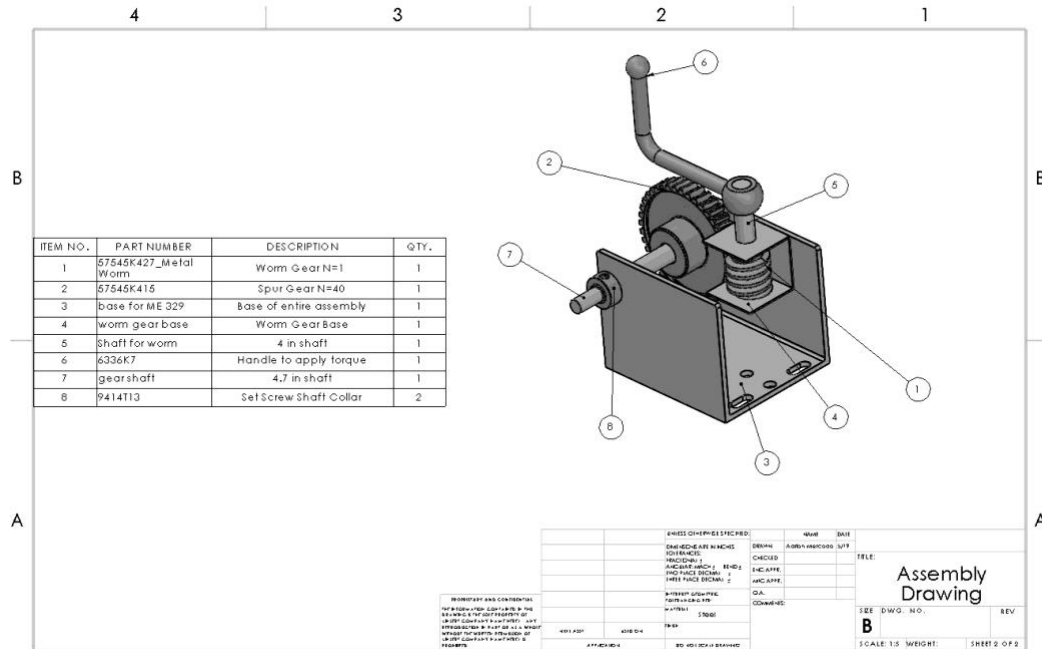
The Gearbox project successfully met its primary objective of designing a functional, low-cost, and compact gearbox capable of supporting significant loads while satisfying the project's design constraints. Analytical calculations and experimental testing demonstrated that the worm gear system provided adequate torque multiplication, self-locking capability, and structural reliability. Although the set screw connection was identified as the governing failure mode, the overall design achieved the highest performance-per-dollar ratio among all groups. Future improvements, including a second set screw and stronger shaft material, would further increase the load capacity and reliability of the system. The project provided valuable experience in mechanical design, manufacturing, analysis, and testing while reinforcing the importance of considering connection design alongside primary structural components.

## **Appendix List:**

Appendix A: Drawings

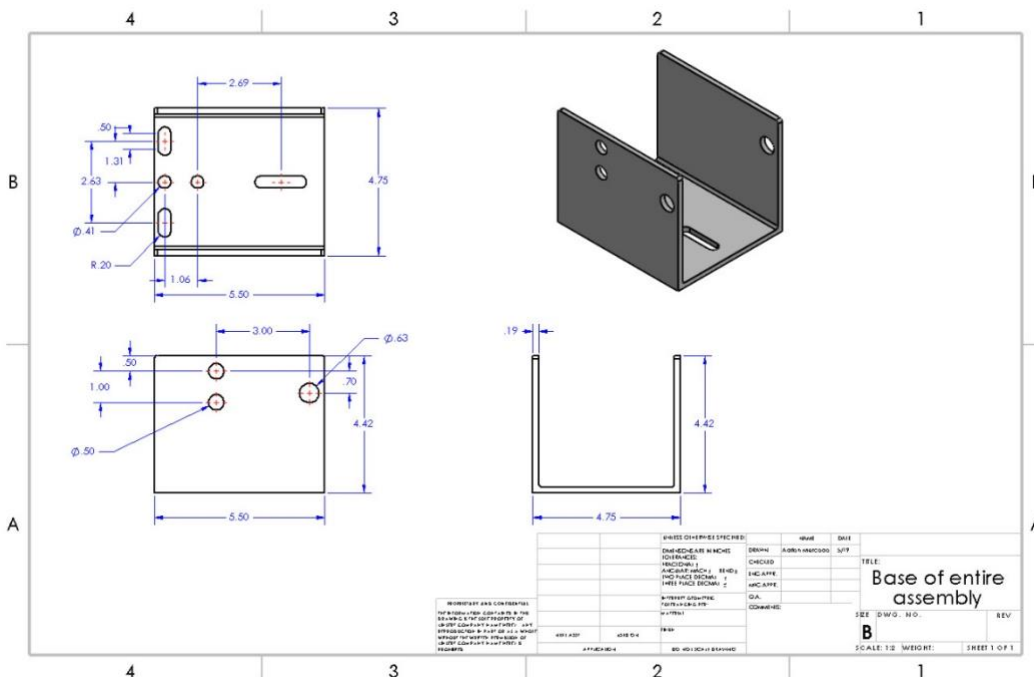
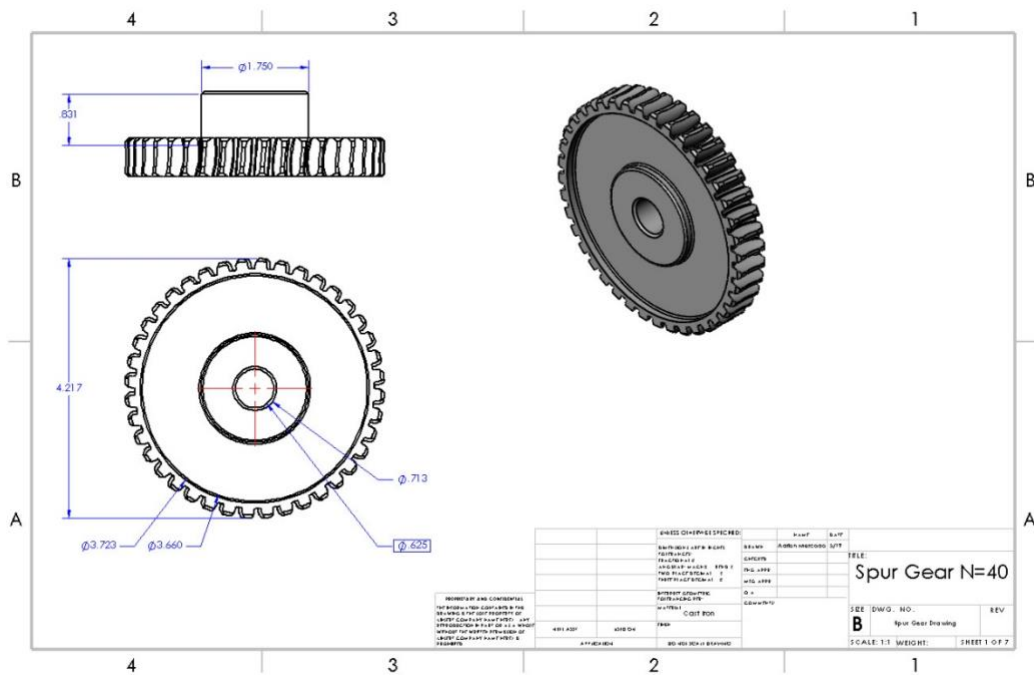
Appendix B: MATLAB codes

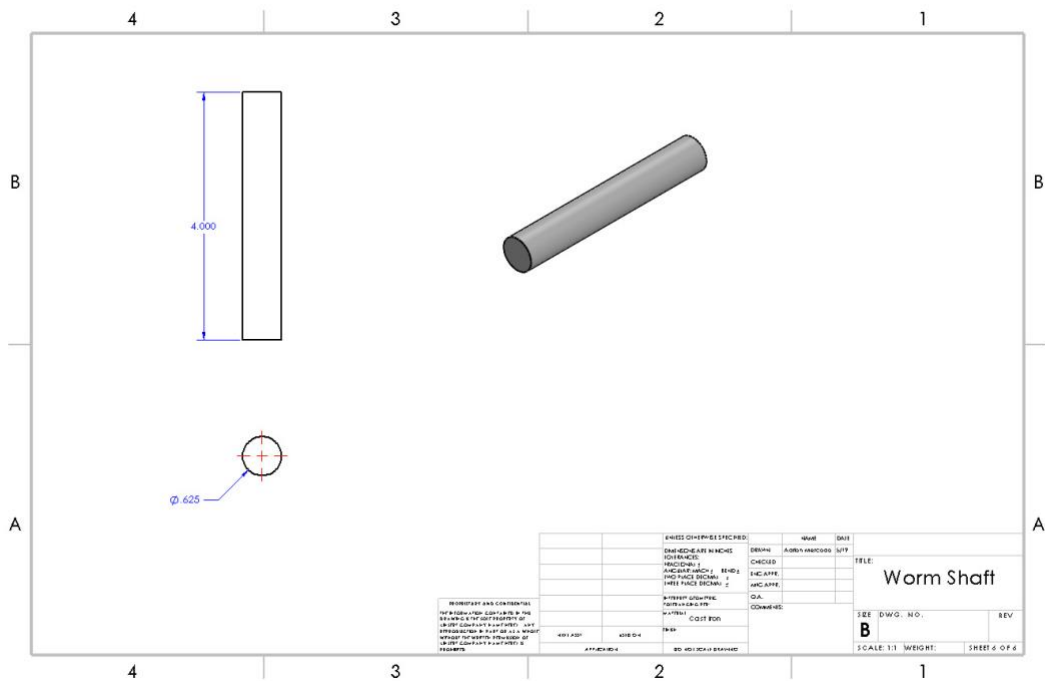
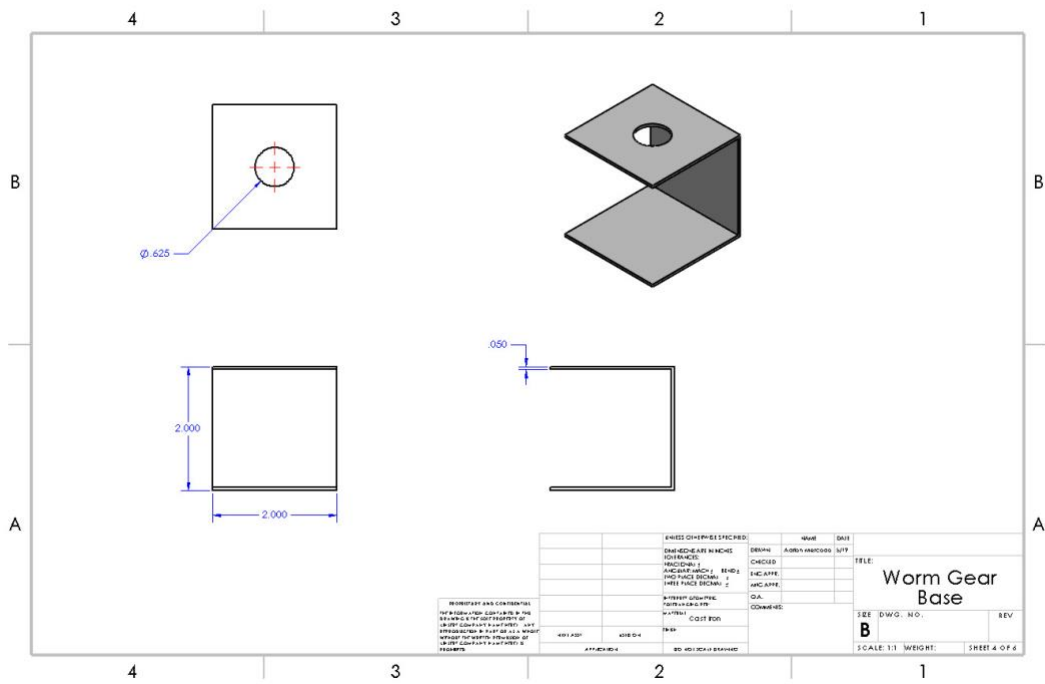
## Attachment A: Drawings

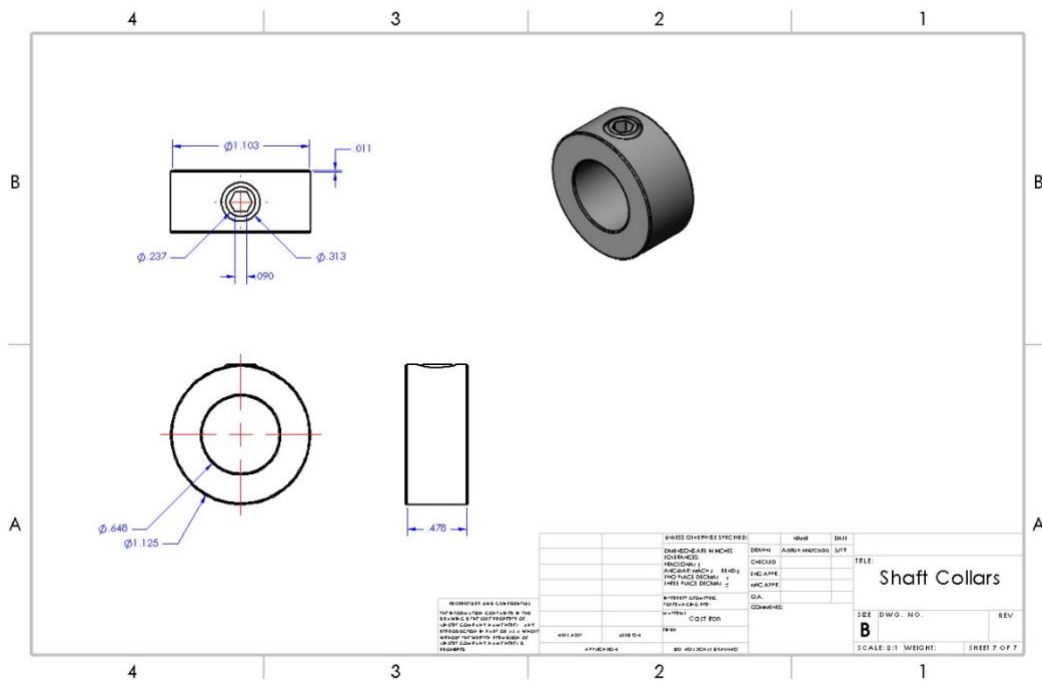
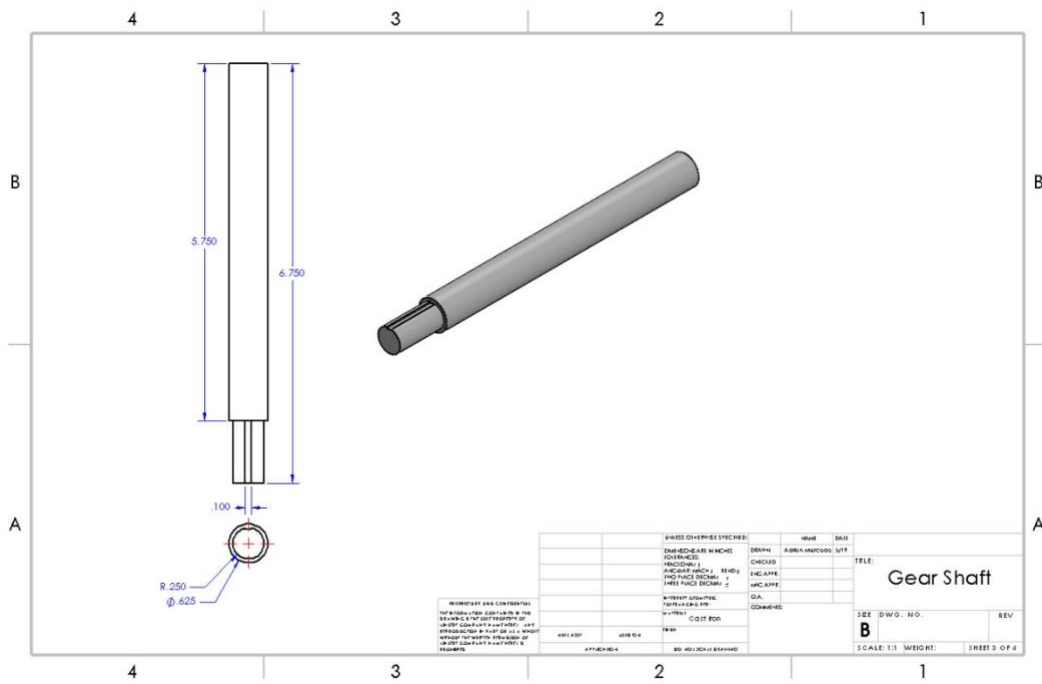




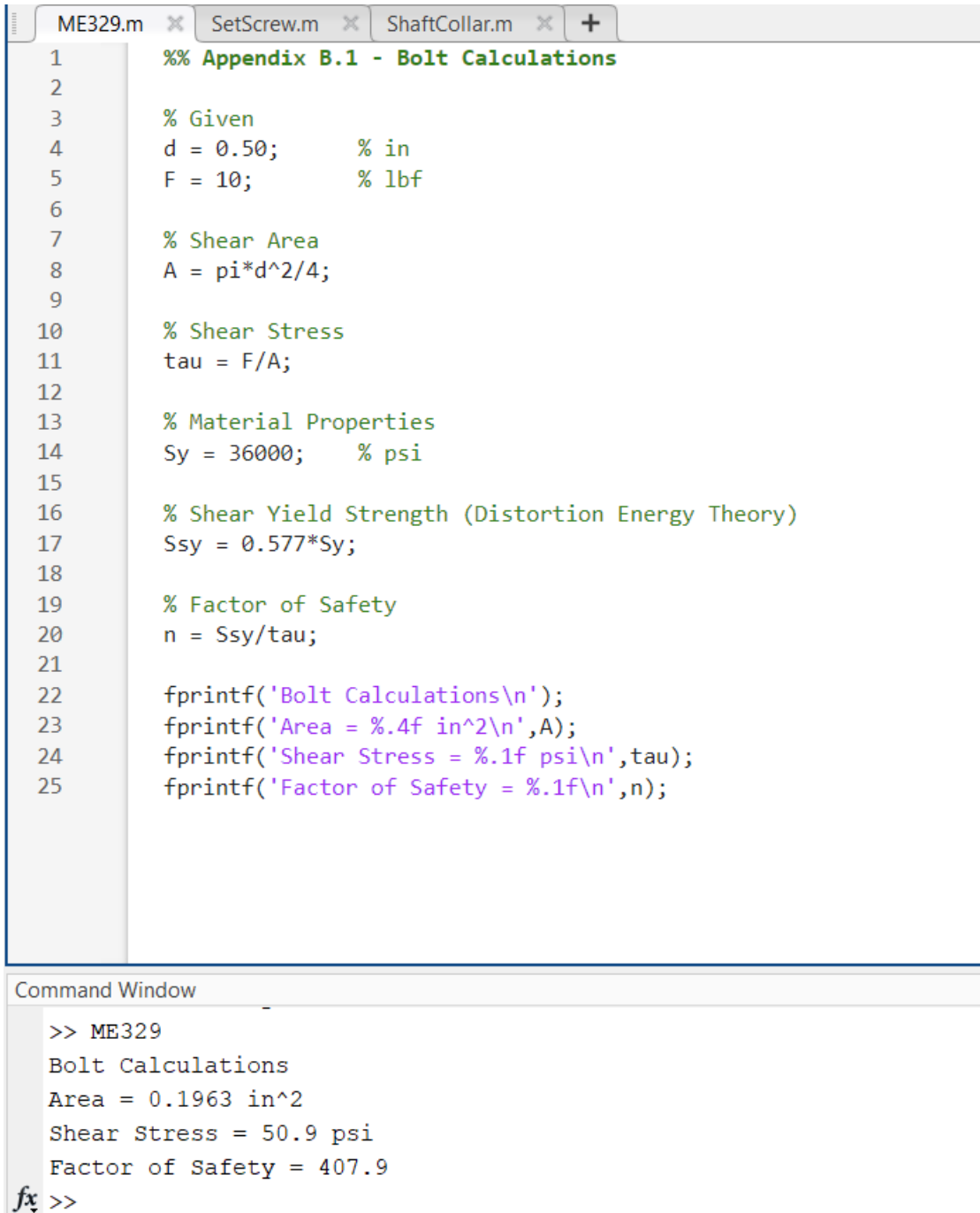
CAL POLY







## Attachment 2: MATLAB Codes



The image shows a MATLAB environment with three open script files: ME329.m, SetScrew.m, and ShaftCollar.m. The ME329.m script contains MATLAB code for bolt calculations. The Command Window shows the execution of the ME329 script, displaying the results of the calculations.

```
1 %% Appendix B.1 - Bolt Calculations
2
3 % Given
4 d = 0.50; % in
5 F = 10; % lbf
6
7 % Shear Area
8 A = pi*d^2/4;
9
10 % Shear Stress
11 tau = F/A;
12
13 % Material Properties
14 Sy = 36000; % psi
15
16 % Shear Yield Strength (Distortion Energy Theory)
17 Ssy = 0.577*Sy;
18
19 % Factor of Safety
20 n = Ssy/tau;
21
22 fprintf('Bolt Calculations\n');
23 fprintf('Area = %.4f in^2\n',A);
24 fprintf('Shear Stress = %.1f psi\n',tau);
25 fprintf('Factor of Safety = %.1f\n',n);
```

Command Window

```
>> ME329
Bolt Calculations
Area = 0.1963 in^2
Shear Stress = 50.9 psi
Factor of Safety = 407.9
fx >>
```



```
1 %% Appendix B.2 - Set Screw Calculations
2 % Given
3 d_shaft = 0.625; % in
4 T = 150; % lb-in
5
6 % Tangential Force
7 r = d_shaft/2;
8 Ft = T/r;
9
10 % Set Screw Properties
11 dr = 0.1887; % in (minor diameter)
12
13 % Shear Area
14 A = pi*dr^2/4;
15
16 % Shear Stress
17 tau = Ft/A;
18
19 % Material Properties
20 Ssy = 20000; % psi
21
22 % Factor of Safety
23 n = Ssy/tau;
24
25 fprintf('Set Screw Calculations\n');
26 fprintf('Tangential Force = %.1f lbf\n',Ft);
27 fprintf('Shear Area = %.4f in^2\n',A);
28 fprintf('Shear Stress = %.0f psi\n',tau);
29 fprintf('Factor of Safety = %.2f\n',n);
```

## Command Window

```
>> SetScrew
Set Screw Calculations
Tangential Force = 480.0 lbf
Shear Area = 0.0280 in^2
Shear Stress = 17164 psi
Factor of Safety = 1.17
fx >>
```



```
ME329.m x SetScrew.m x ShaftCollar.m x +
1      %% Appendix B.3 - Shaft Collar Calculations
2
3      % Given
4      P = 300;           % lbf
5      N = 2;           % number of collars
6
7      % Load per Collar
8      F = P/N;
9
10     % Manufacturer Rating
11     F_allow = 1000;    % lbf
12
13     % Factor of Safety
14     n = F_allow/F;
15
16     fprintf('Shaft Collar Calculations\n');
17     fprintf('Load per Collar = %.1f lbf\n',F);
18     fprintf('Factor of Safety = %.2f\n',n);
```

---

Command Window

```
>> ShaftCollar
Shaft Collar Calculations
Load per Collar = 150.0 lbf
Factor of Safety = 6.67
>> ShaftCollar
Shaft Collar Calculations
Load per Collar = 150.0 lbf
Factor of Safety = 6.67
fx >>
```

```

Lecture_2_Shift_Example.m x worm.m x +
C:\Users\Eric\Downloads\Lecture_2_Shift_Example.m

1
2 % Compact Shaft Fatigue Check - Keyway + Shoulder Step
3 clc; clear; close all
4
5 % 1. GIVENS
6 Ma = 62.5; % alternating bending moment, lb-in
7 Tm = 150; % mean torque, lb-in
8 Sut = 45; % ksi
9 Sy = 25; % ksi
10 E = 30e6; % Modulus of Elasticity for Steel (psi)
11 n_design = 1.5;
12
13 % 2. SHAFT GEOMETRY
14 d = 0.500; % small shaft diameter, in
15 D = 0.625; % large shaft diameter, in
16 r = 0.030; % shoulder fillet radius, in
17 L_eff = 4.0; % EFFECTIVE LENGTH (in)
18 fprintf('D/d = %.3f\n', D/d)
19 fprintf('r/d = %.3f\n\n', r/d)
20
21 % 3. STIFFNESS ANALYSIS (New Section)
22 % Based on Cantilever -
23 I = (pi * d^4) / 64;
24
25 % Since M = F*L at the support, we can find the equivalent For
26 F_equiv = Ma / L_eff;
27
28 % Max Deflection calculation: y_max = (F * L^3) / (3 * E * I)
29 y_max = (F_equiv * L_eff^3) / (3 * E * I);
30
31 % Slope at end (Theta)
32 theta_max = (F_equiv * L_eff^2) / (2 * E * I);
33
34 fprintf('Max Deflection (y_max): %.6f in\n', y_max)
35 fprintf('Slope at End (theta): %.6f rad\n\n', theta_max)
36
37 % 4. ENDURANCE LIMIT MODIFIERS
38 Se_p = 0.5*Sut; % ksi
39 ka = 2.7*Sut^(-0.265); % machined surface
40 kb = 0.879*d^(-0.107); % size factor
41 kc = 1; % loading factor
42 kd = 1; % temperature factor
43 ke = 0.753; % reliability factor
44
Command Window

D/d = 1.250
r/d = 0.060

Max Deflection (y_max): 0.003622 in
Slope at End (theta): 0.001358 rad

Corrected endurance limit Se = 15.792 ksi

--- Fatigue Check at d = 0.500 in ---
Governing fatigue safety factor = 0.888
Fatigue result: FAIL

Governing yield safety factor = 1.585

```

```

Lecture_2_Shift_Example.m x worm.m x +
C:\Users\Eric\Downloads\worm.m
1 %
2 % ME 329 Worm Parametric Tool -
3 clear; clc;
4
5 % --- 1. WINCH PARAMETERS ---
6 P_load = 17.5; % Max testing load (lbf)
7 L_arm = 12; % Moment arm length (in)
8 N_w = 1; % Single start worm
9 N_g = 40; % 40:1 Gear ratio
10 p_d = 7.5; % Diametral pitch
11 alpha_n = 20; % Normal pressure angle (deg)
12 mu = 0.15; % Friction (Steel-on-Steel)
13 d_w = 1.25; % worm diameter
14 r_handle = 5.0; % Standard winch handle length
15 face_width = 0.50; % Effective contact width (in)
16
17 % Material: Hardened Alloy Steel
18 S_y = 150000; % Yield Strength (psi)
19 S_es = 150000; % Surface Endurance Limit (psi)
20 n_safety = 2.0; % ME 329 Required Safety Factor
21
22 % --- 2. KINEMATICS & FORCES ---
23 d_g = N_g / p_d;
24 T_out = P_load * L_arm;
25 W_tg = T_out / (d_g / 2);
26 lead = N_w * (pi / p_d);
27 lambda = atand(lead / (pi * d_w)); % Lead angle
28
29 % --- 3. EFFICIENCY & SELF-LOCKING ---
30 % Efficiency calculation (Forward drive)
31 eff = (cosd(alpha_n) - mu*tand(lambda)) / (cosd(alpha_n) + mu*
32
33 % Self-locking Check
34 % A worm set is self-locking if mu > cos(alpha_n) * tan(lambda)
35 mu_static_req = cosd(alpha_n) * tand(lambda);
36 is_self_locking = mu > mu_static_req;
37
38 % Input torque and force required at the handle
39 T_in = (W_tg * (d_g/2)) / ((N_g/N_w) * eff);
40 F_input = T_in / r_handle;
41
42 % --- 4. STRESS ANALYSIS ---
43 % Bending Stress (Lewis)
44
Command Window
--- ME 329 Winch Analysis: 17.5 lb Load ---
Lead Angle (lambda): 6.09 degrees
System Efficiency: 39.37%
Self-Locking: Yes
-----
User Input Force: 2.67 lbf (Guideline: <10 lbf)
Bending Stress: 3375 psi
Contact Stress: 29835 psi
Bending: PASS (SF = 44.4)
Surface Durability: PASS (SF = 5.0)

```